FEATURE GUIDE · SMART INDIA HACKATHON SIH26047 · PATIENT CASE-TAKING SOFTWARE

OPD Saathi

A multilingual, voice-first check-in kiosk and live queue for crowded government OPDs — the patient's history is ready before they reach the doctor.

This guide explains every part of the system in plain words: what each screen is for, what it shows, how the pieces connect, how to install it at a hospital and run it day to day, and what to say about each one in a presentation. Every screenshot comes from the running application with fictional demo patients.

Hindi · Marathi · English | Tap, speak or type | Live queue + QR status | Fair triage score | English case sheet for doctors | Reports & administration | Works offline (short form) | History only — no diagnosis | One setup per hospital | Many kiosks, one queue

Updated 11 October 2026: adds setting up a new hospital, step-by-step administration (departments, staff, kiosks), kiosk setup links and re-activation, delete, a full command reference and troubleshooting. Synthetic data only.

START HERE

The big picture

Government OPDs see hundreds of patients a morning. Patients speak different languages, describe problems in their own way, and wait without knowing their turn. Doctors get a few minutes per patient and spend much of it collecting basic history.

What OPD Saathi does

- Collects the patient's history **before** the consultation, in their language, at a kiosk.
- Spots warning signs and gives the patient a **fair place in the queue**.
- Gives the doctor a **one-page English case sheet** with the patient's own words alongside.
- Shows staff a **live queue** and patients a **live status page** on their phone.
- Gives administrators **reports** and control over staff, kiosks and departments.

What it deliberately does not do

- It never diagnoses, names a disease, suggests medicines or gives treatment advice.
- It never shows the patient their urgency score — only staff see it.
- AI can only *add* caution; safety rules decide, and staff confirm.
- Clinical questions and rules are drafts until clinicians review them.

One patient's journey

Kiosk Language, consent, age/sex, danger signs, 8–15 questions → **Token + QR** Department, place in line, phone status page → **Nurse queue** Live list, alerts, fair calling order → **Doctor** English case sheet, transcript, edit, PDF → **Admin** Reports, staff, kiosks, departments

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1 · ROLES

Who uses what

Five kinds of users, one web application. Staff screens change with the signed-in role; patients never need an account.

WHO	WHERE	WHAT THEY DO THERE
Patient	Kiosk `/kiosk`	Chooses Hindi, Marathi or English; agrees to consent; gives age and sex; flags danger signs; answers 8–15 questions by tapping, speaking or typing; checks a read-back of their history; receives a token, department, place in line and a QR code.
Patient's phone	QR → `/s/<token>`	Live place in the queue, estimated wait, and progress (Waiting → Called → With doctor → Done) in their language. No app or login needed.
Nurse / reception	Staff → **Queue**, **Patients**	Watches the live queue and alerts, calls patients, marks no-shows, changes priority with a reason, confirms uncertain departments, registers patients who need help (assisted registration).
Doctor	Staff → **Queue**, **Consultations**	Opens the case sheet, reviews red flags, the AI summary, full history and the patient's own words; corrects fields; starts and finishes the consultation; downloads or prints the case sheet PDF.
Administrator	Staff → **Reports**, **Settings** (and all other pages)	Reads activity reports and exports CSV; creates, turns off and deletes staff accounts, kiosks and departments; sets up each kiosk with a one-time setup link; sets the hospital-wide "time saved" estimate.

Staff sign-in and the shell

Staff sign in with hospital, email and password. The left sidebar (Queue, Consultations, Patients, Reports, Settings) shows only what the role may open — a nurse or doctor who tries to open Settings is sent back to the Queue. The header shows the signed-in person, their role and a **Live** connection badge. On phones the sidebar becomes a top bar.

3
languages
Hindi, Marathi and English everywhere, with about 800 translated staff phrases and a full kiosk translation.

5
staff pages
Queue, Consultations, Patients, Reports, Settings — all redesigned for laptops, tablets and phones.

8
demo departments
General Medicine, Paediatrics, Orthopaedics, Obstetrics & Gynaecology, ENT, Dermatology, Ophthalmology, General Surgery — every new hospital starts with these too.

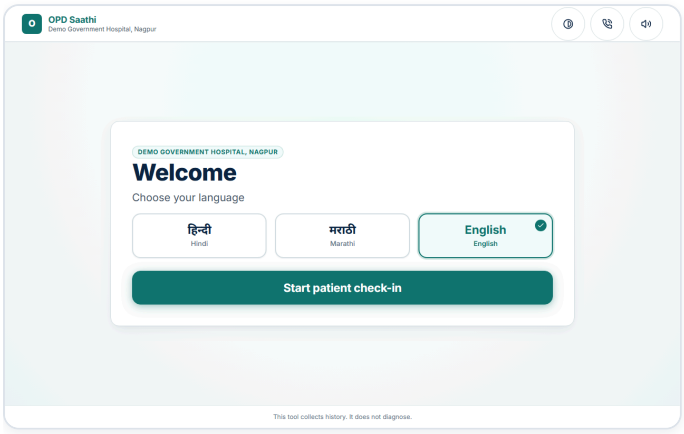
SAY IT IN THE PPT

One system, five audiences: the patient checks in at a kiosk in their own language, follows their turn on their phone, nurses run a live fair queue, doctors get an English case sheet, and administrators see the whole OPD.

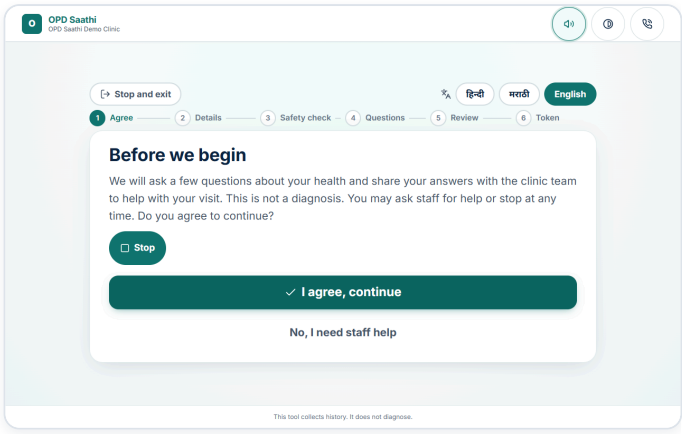
2 · PATIENT KIOSK

Check-in, step by step

A touch-screen kiosk (any tablet or PC with Chrome or Edge). Six clear steps are shown at the top of every screen: Agree → Details → Safety check → Questions → Review → Token.

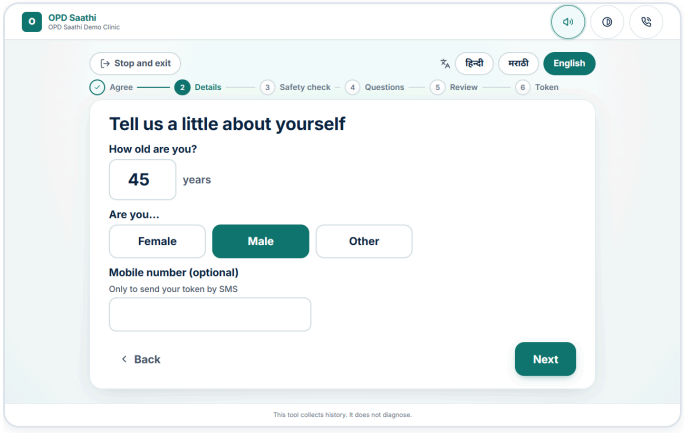


Welcome: pick a language, then "Start patient check-in". Comfort buttons sit top-right.

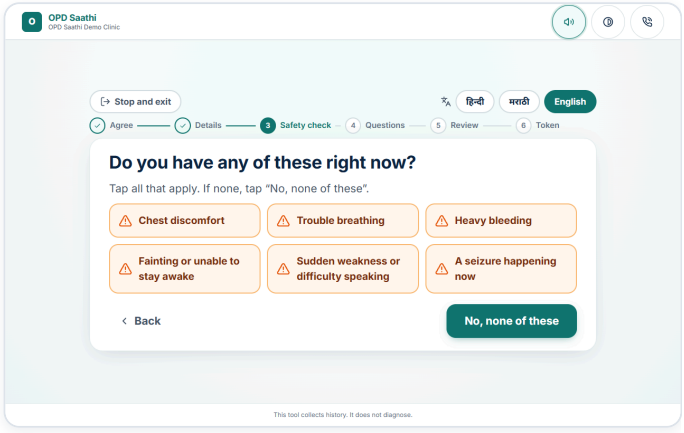


Agree: the consent is read aloud. "No, I need staff help" is always offered.

1. **Language.** हिन्दी, मराठी or English. The patient can switch language on any later screen too.
2. **Consent.** A short notice in plain words, spoken automatically. Nothing is collected before the patient agrees.
3. **Details.** Only age and sex are needed. A mobile number is optional and only used, if the hospital turns it on, for SMS queue updates.



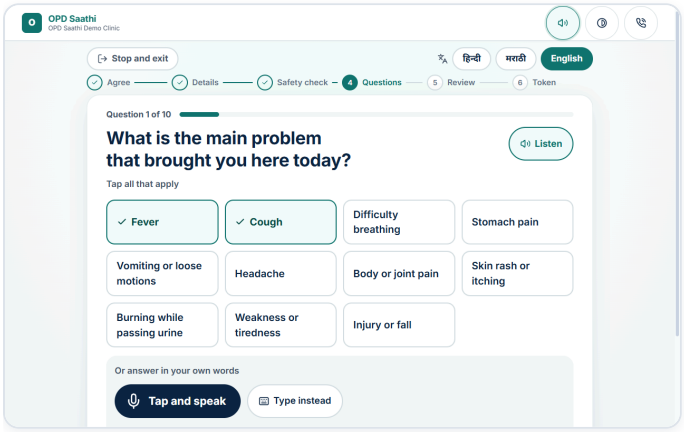
Details: just age and sex.



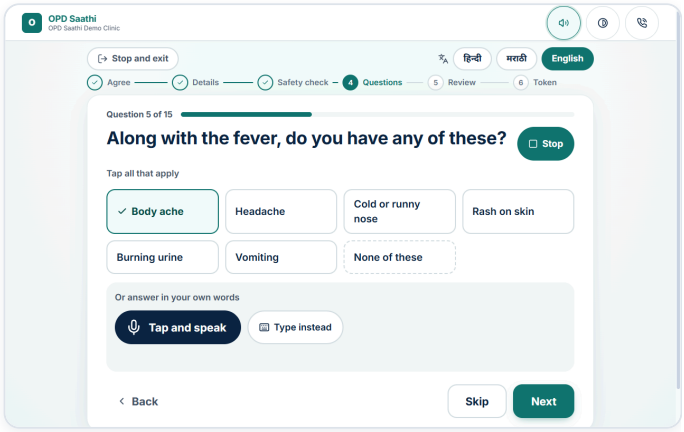
Safety check: six large danger-sign tiles come *before* any questions.

1. **Safety check.** Chest discomfort, trouble breathing, heavy bleeding, fainting, sudden weakness or difficulty speaking, a seizure happening now. Tapping one sends an **urgent** alert to staff at once, so a serious patient is noticed within seconds instead of after a full interview.

Smart questions, read-back and token

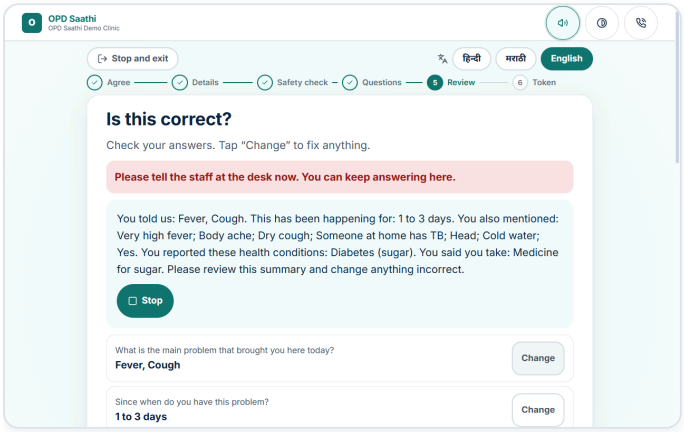


Questions: tap one or more answers, or "Tap and speak", or "Type instead". Each question can be listened to.

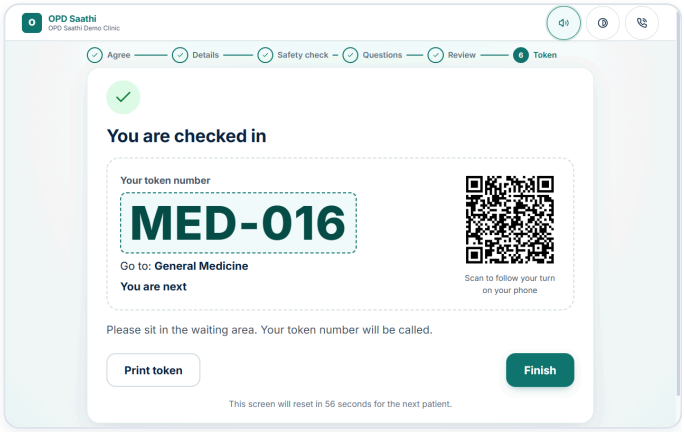


Follow-ups adapt: choosing fever and cough brings fever and breathing questions.

1. **Questions (8–15).** Built from 13 topic modules (fever, breathing, stomach, pain, urine, skin, injury, sleep/energy, past illness, medicines, allergies, family history, anything else). The interview follows the complaint and skips what does not apply — for example, no pregnancy or period questions for men or children. A progress bar shows how many are left. **Back** changes the previous answer; **Stop and exit** ends the visit safely at any time.
2. **Review.** The whole history is read back in the patient's language. Any answer can be changed before confirming "Yes, this is correct". Confirmation is recorded — doctors can see it.
3. **Token.** A large token (for example MED-016), the department to go to, the place in line, and a QR code for the phone. "Print token" prints a slip. The screen resets itself after about a minute for the next patient.



Review: the read-back, with a "Change" button beside every answer.



Token: department, place in line and a QR code for the phone status page.

SAY IT IN THE PPT

Most answers are a single tap, so even patients who cannot read or type can finish in a few minutes. The patient hears and confirms their own history before it reaches the doctor.

3 · KIOSK COMFORT, SAFETY AND OFFLINE MODE

Made for every patient

Comfort and accessibility

- **Voice guidance** reads every screen aloud; **Listen** repeats a question. Turn it off with one tap.
- **Speak your answer** in Hindi, Marathi or English; the words appear for checking before they are used.
- **High contrast** mode for low vision; every button is at least 64 px — easy for older hands.
- **Call staff** button on every screen raises a "Patient asked for help" alert in the staff queue.
- **Back** and **Stop and exit** on every step; idle kiosks reset themselves so the next patient never sees someone else's answers.

Safety on the kiosk

- Danger signs are asked **first** and alert staff immediately.
- Answers that describe a warning sign (for example "very high fever", "blood in phlegm") raise a red flag through safety rules — in every language.
- The patient only ever sees neutral wording; no scores, no disease names.
- An anonymous "emergency" button was deliberately removed because anyone could trigger false alarms; real emergencies are handled by staff in person.

Old prescription scan AI

On the medicines question the patient can photograph an old prescription; the medicines are read automatically and shown back for the patient to confirm. (Needs the Gemini key.)

Offline short form

If the internet drops, the kiosk offers a short form (complaint, duration, severity, conditions, medicines, allergies, danger signs). It is **encrypted on the kiosk**, gets a provisional token, and is sent automatically when the connection returns; staff then confirm the real token. Sending twice can never create two visits.

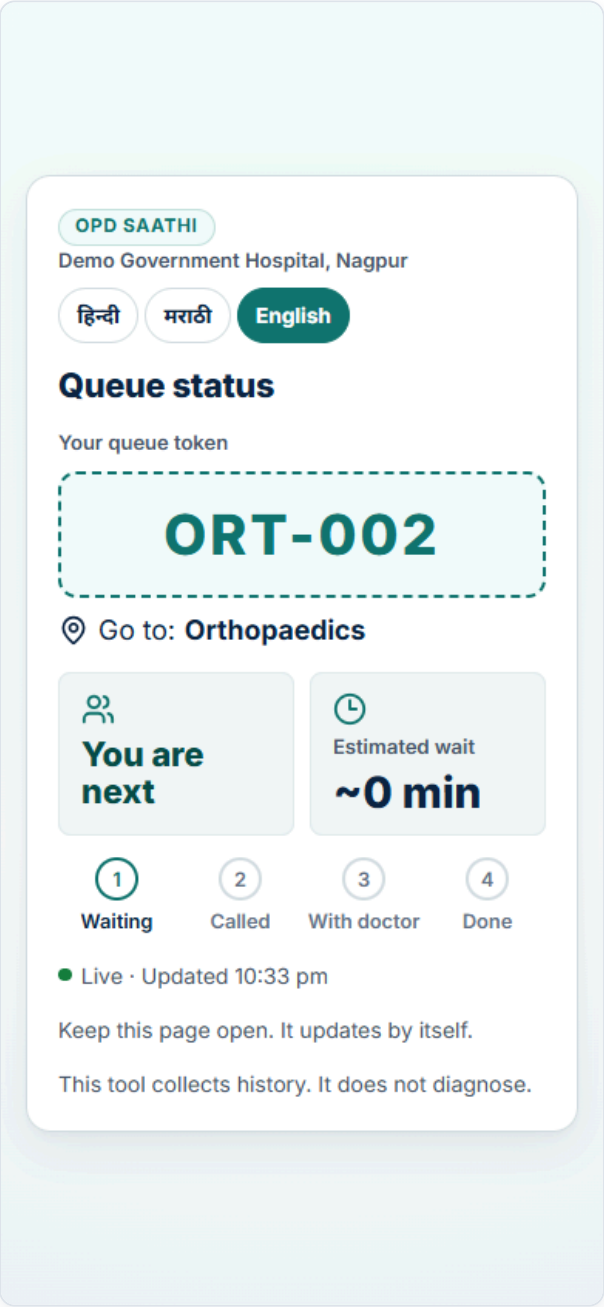
Kiosk setup in one minute

An administrator registers a kiosk in Settings and gets a **one-time activation code** with a **Copy setup link** button. Opening that link once on the kiosk device activates it. From then on the kiosk **stays signed in by itself** — overnight, after a restart, for as long as the administrator keeps it in service — and shows the hospital's name at the top. It reports that it is online about once a minute, so Settings shows it as Online or Offline. If the device is replaced, the administrator issues a **new activation code for the same kiosk**; its history is kept.

> **SAY IT IN THE PPT**
> Accessibility is built in, not added on: voice in three languages, high contrast, very large buttons, a call-staff button on every screen, and an offline mode so check-in never stops.

4 · PATIENT PHONE STATUS PAGE

"When is my turn?" — answered on the phone



Scanned from the token QR. Here a knee-pain patient was routed to Orthopaedics (ORT-002).

The QR code on the token opens a page on the patient's own phone — no app, no login, no personal details shown.

What it shows

- The token and the department to go to.
- Place in line (for example "You are next") and an estimated wait.
- A four-step progress track: **Waiting → Called → With doctor → Done**.
- Language switch (Hindi, Marathi, English).
- It updates by itself as the queue moves.

Why it matters

Patients can sit down, step outside or look after a child instead of crowding the doctor's door. The link shows only that one visit and contains no name or health details.

Optional SMS updates

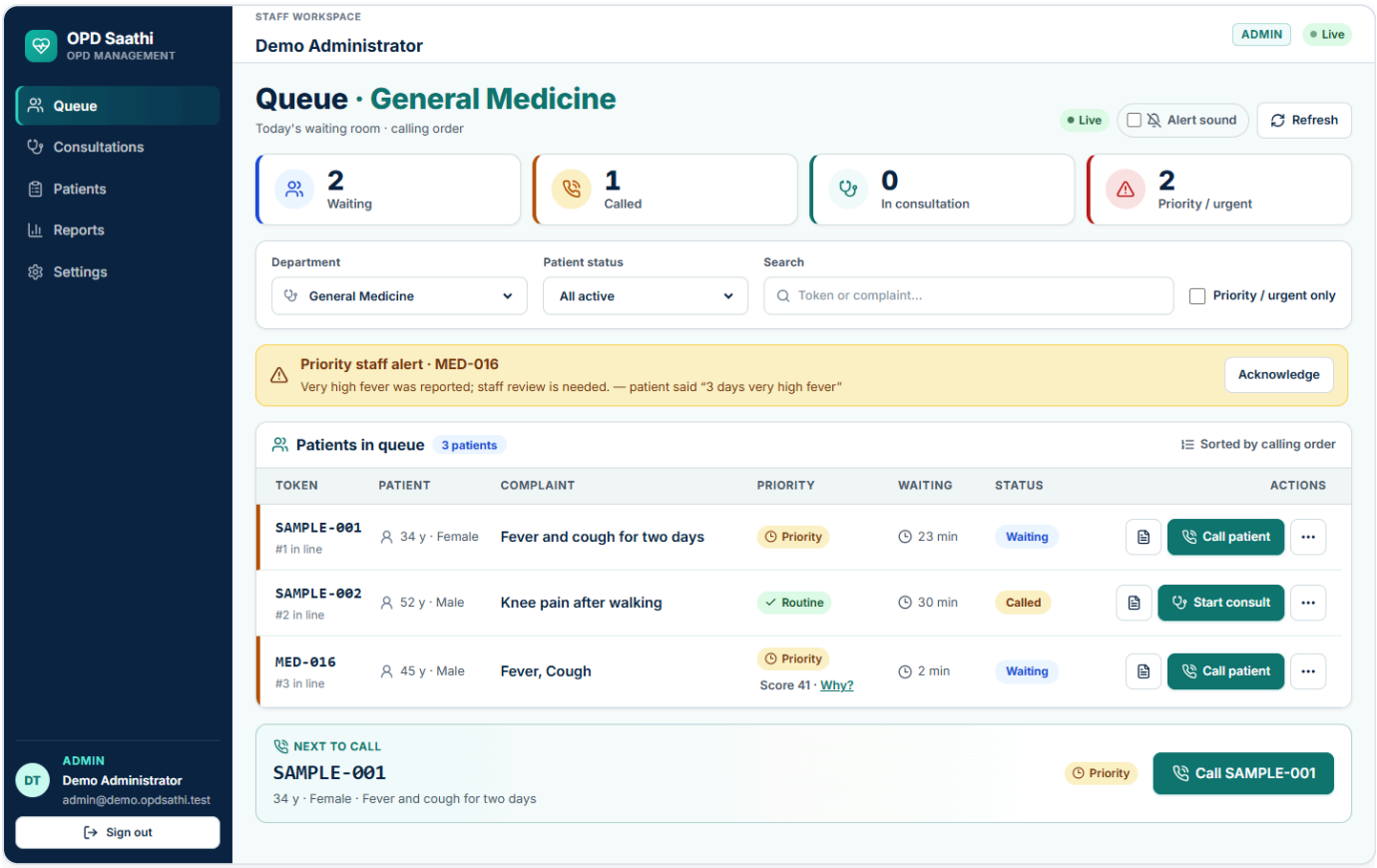
If the hospital connects an SMS provider and the patient gives a number and opts in, they receive their token and queue position by SMS (no health details). It is off by default.

> **SAY IT IN THE PPT**
> The patient leaves the kiosk with a token and a live link — they always know where they stand, which calms the waiting area and reduces crowding at the door.

5 · STAFF · QUEUE

The live queue dashboard

The home page for nurses and doctors: today's waiting room for one department, in fair calling order, updating live.



Queue for General Medicine: counts, filters, a priority alert, the patient table and the "Next to call" panel.

What is on the page

- **Four counters** — Waiting, Called, In consultation, Priority/urgent. Click one to filter.
- **Filters** — department, patient status, search by token or complaint, "priority/urgent only".
- **Alerts** — red flags, "Patient asked for help" and "AI suggests this patient may be more urgent (please confirm)", each with the reason and the patient's own words, and an **Acknowledge** button. Optional alert sound.
- **Patients in queue** — token and place, age/sex, complaint (translated to English), priority with score and "Why?", waiting time, status.
- **Next to call** — who should be called next, with a one-click Call.

Actions

- **Call patient → Start consult → Complete** (the main button changes with the patient's state).
- **Open case sheet** (document icon).
- **⋯ menu**: mark no-show; **change priority** (requires a written reason, recorded for audit).
- Confirm the department when the automatic routing was unsure.

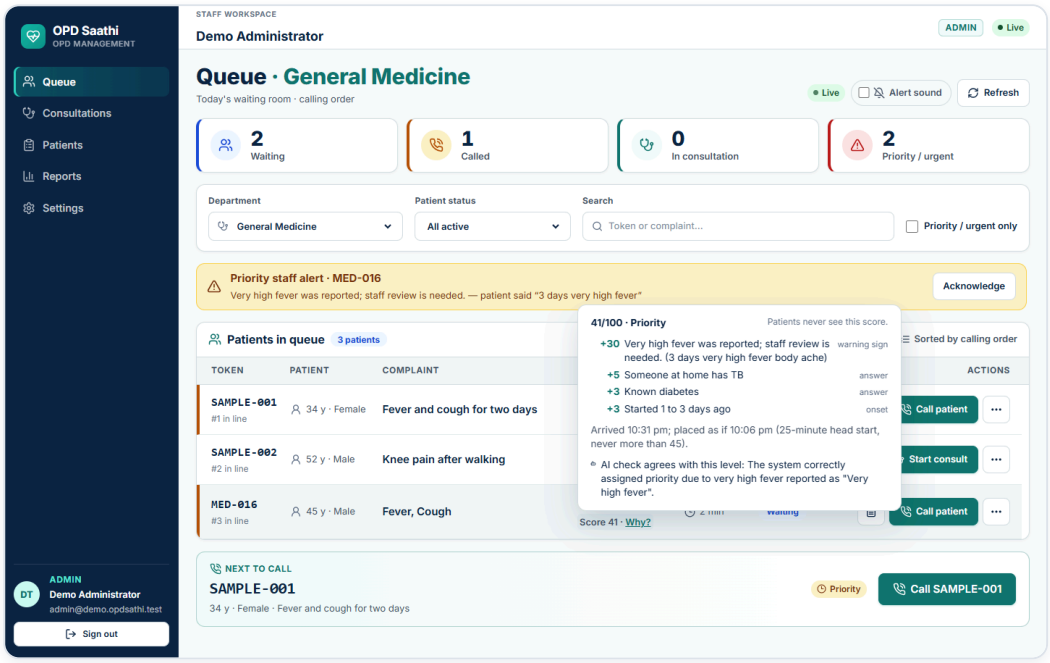
Live and reliable

- Updates arrive instantly over a live connection; a badge shows Live or Reconnecting.
- Handles very long queues: a 5,000-patient list applies an urgent change in well under 100 ms.
- Unfinished visits from earlier days are closed automatically, so every morning starts with a clean queue.

6 · FAIR QUEUE

How the triage score and calling order work

The question every OPD asks: "Why was that person called before me?" OPD Saathi answers it with a transparent score that staff can open and read line by line.



"Why?" opens the explanation: 41/100 · Priority — every point with its reason, how much head start it gave, and the AI check.

The score (0–100) — staff only

- Built from objective signs: an urgent red flag or danger tile is worth 70, a priority flag 30; warning answers add more (blood in phlegm +25, very little urine +10; at most 25 together); vulnerability adds points (infant under 1 year +20, aged 75+ +12, pregnant +12); a problem that started today +6.
- The patient's own 0–10 rating is worth at most 4 of 100 points — typing "10" cannot jump the queue.
- Levels: **Routine** below 30, **Priority** 30+, **Urgent** 70+ or any urgent red flag.
- Patients never see the score.

The calling order

- **Urgent** patients go to the front.
- Everyone else keeps arrival order, but higher scores get a **head start**: 0.6 minutes per point, never more than 45 minutes. So a routine patient can only be passed by someone who arrived less than 45 minutes after them — no one waits forever.
- Staff can override priority, but must type a reason; the change and reason are kept.
- **AI is a checker, not a decider** AI : after check-in an AI review may raise an "AI suggests… please confirm" alert; it never lowers urgency, and an AI flag alone counts as priority at most.
- All weights and red-flag rules live in readable files marked "draft — needs clinician review", so doctors can check every number.

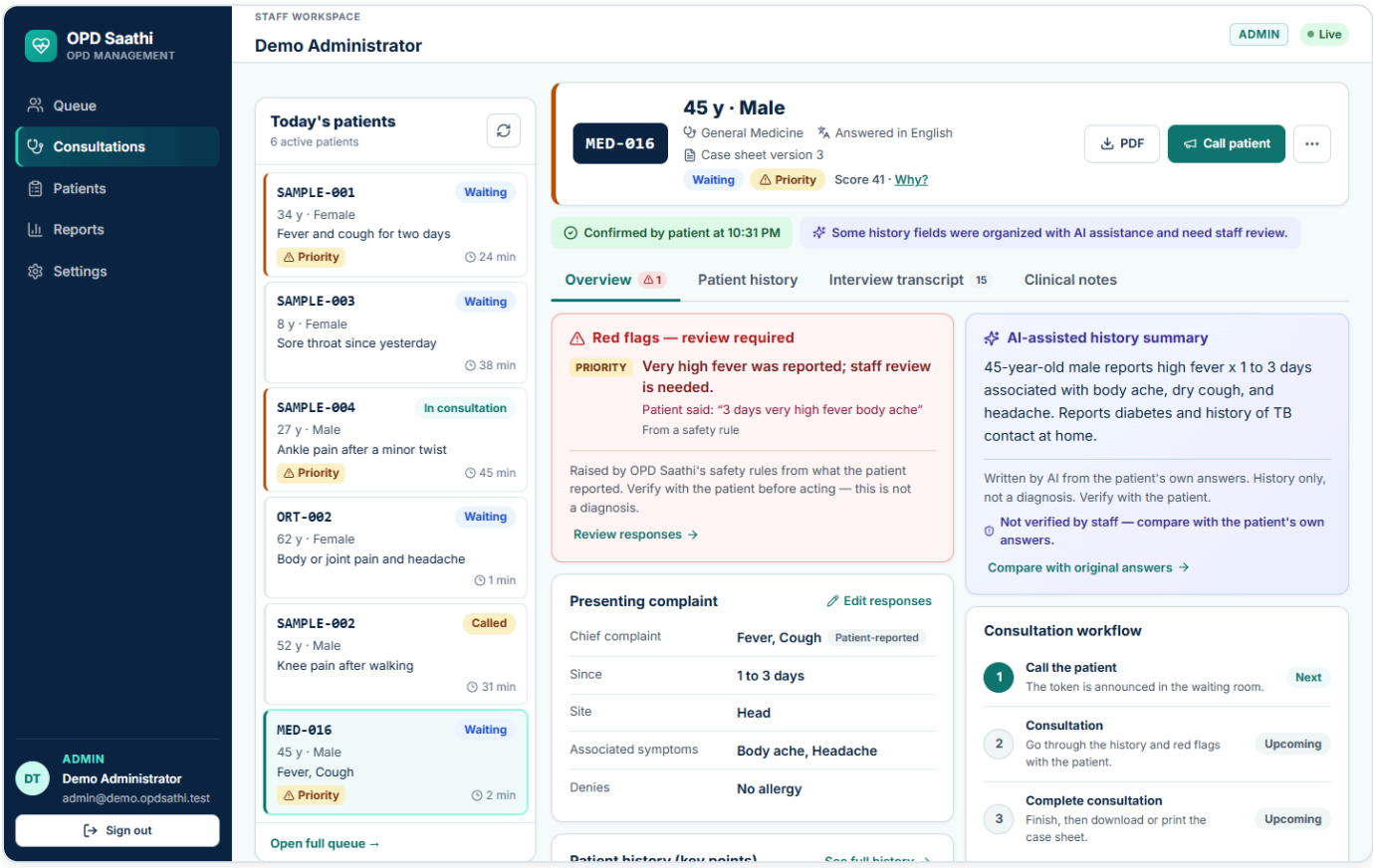
SAY IT IN THE PPT

Our queue is fair and explainable: the sickest go first, everyone else keeps their place, a typed "10/10" cannot cheat the system, and any staff member can click "Why?" to see exactly how a position was decided.

7 · STAFF · CONSULTATIONS

The doctor's case sheet

Left: today's patients. Right: one patient, organised into four tabs so the doctor sees what matters first instead of everything at once.



Overview tab: red flags first, the AI-assisted summary (marked "not verified by staff"), the presenting complaint and the consultation workflow.

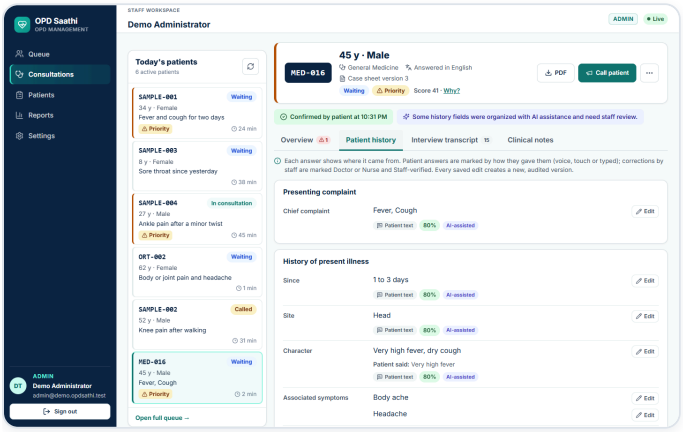
Header (always visible)

- Token, age, sex, department, language answered in, case-sheet version.
- Status, priority and score with "Why?".
- One main action: Call → Start consultation → Finish; plus PDF and a ⋯ menu (print, keyboard shortcuts N, E, P).
- Green "Confirmed by patient at 10:31" and a separate violet AI notice — patient confirmation never counts as staff checking the AI.

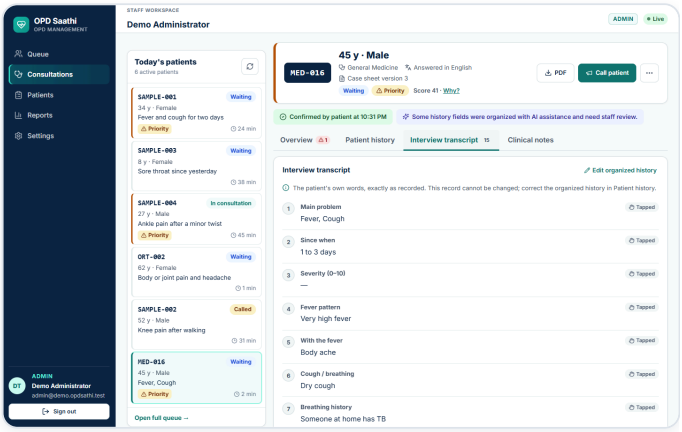
Overview tab

- **Red flags — review required** with the reported words, where the flag came from, and "this is not a diagnosis".
- AI-assisted history summary with "Compare with original answers".
- Presenting complaint, key history points, and the call → consult → complete workflow from the visit's real status.

History, transcript and notes



Patient history: every field, editable, with where each answer came from.



Interview transcript: the patient's exact words, numbered, with how they answered.

Patient history tab

- Complaint, history of present illness, past illness, medicines, allergies, family history, tobacco/alcohol, pregnancy.
- Each field shows its source (voice, tap, typed, nurse, doctor), confidence and an "AI-assisted" tag where relevant, plus the patient's original words.
- **Edit** any field. Every save creates a new version; staff edits are marked **Staff-verified**.
- If two people edit at once, the second is told and can reload or re-apply their change — nothing is silently overwritten.
- Leaving with an unsaved edit asks first.

Transcript and notes tabs

- **Interview transcript**: every question and the full answer in the patient's language, voice/tap/typed, speech confidence, and "AI follow-up" labels on questions the AI chose. This record cannot be changed.
- **Clinical notes**: staff edits on this case sheet, the version and audit note, and corrections the patient made before confirming. (Exam findings and plan are not stored yet — see limitations.)
- **PDF / Print** always produce the full case sheet, whichever tab is open.

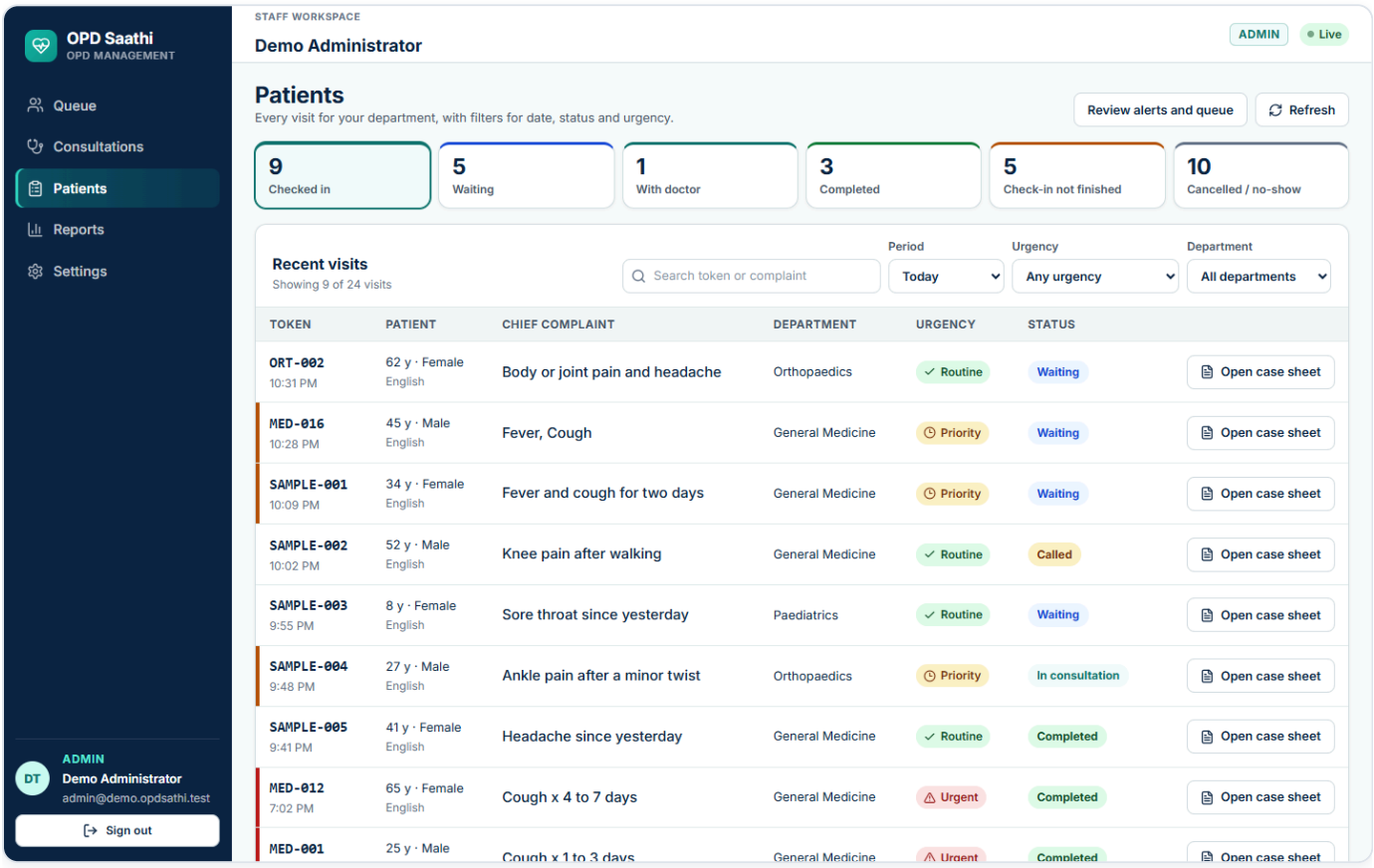
SAY IT IN THE PPT

The doctor gets a one-page English case sheet in seconds — urgent warnings first, an AI brief clearly marked as unverified, and the patient's own Hindi or Marathi words one click away. Every correction is versioned and audited.

8 · STAFF · PATIENTS

The patients register

Every visit in a period — not just today's waiting room — for finding a patient, checking what happened, and registering people who need help.



Status tiles, filters and the visits table, with "Open case sheet" on each row.

What it shows

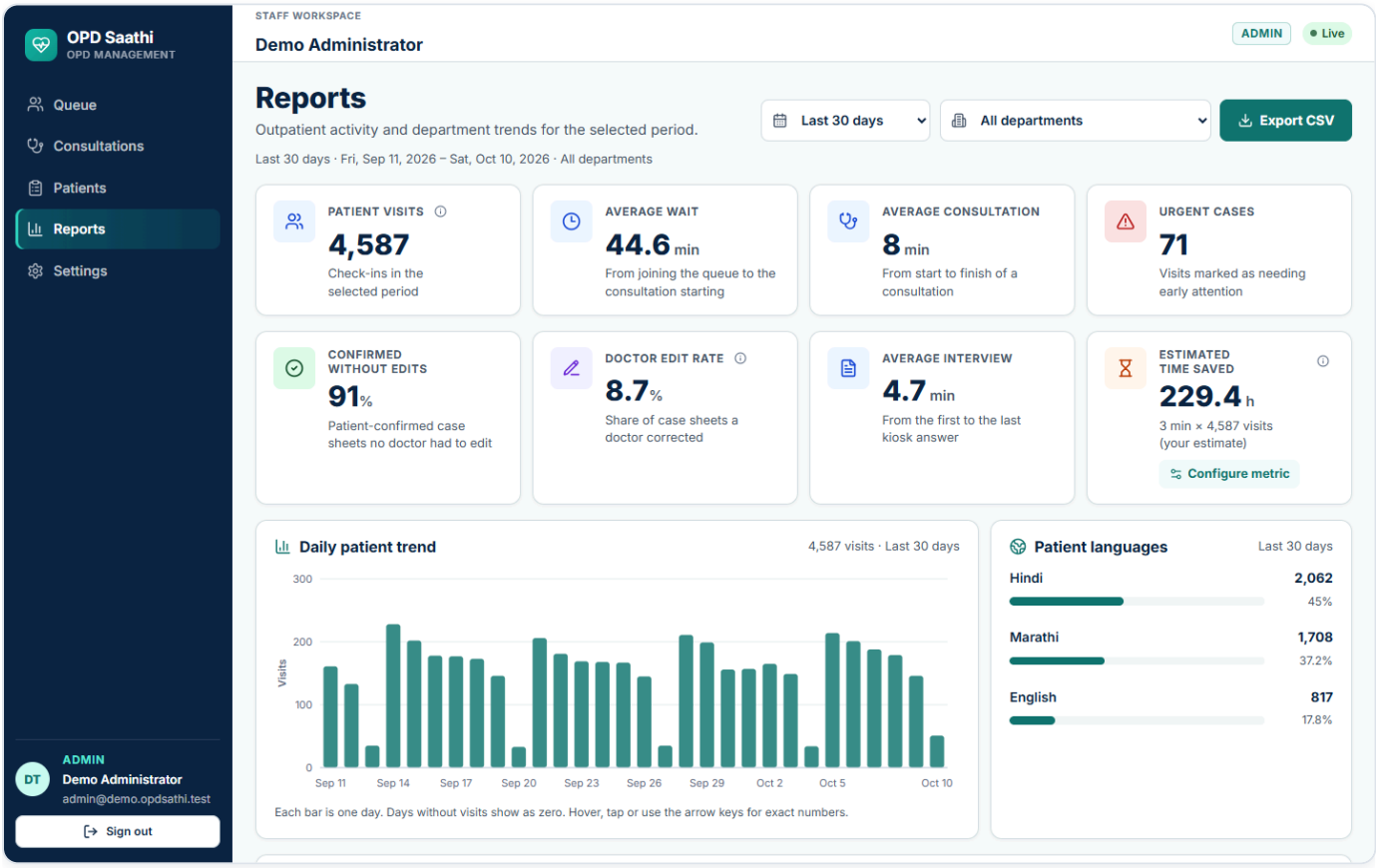
- **Status tiles**: Checked in, Waiting, With doctor, Completed, Check-in not finished, Cancelled/no-show — click to filter.
- Filters for period (today, 7 days, 30 days), urgency and department, and a search box.
- Each visit: token, patient, complaint, urgency, status, language, and a link to the case sheet.

Assisted registration

- A nurse can start a check-in for a patient who cannot use the kiosk (elderly, unwell, unable to read).
- Consent is always shown first — staff cannot skip it.
- The visit then follows the same interview, safety rules and queue as a kiosk check-in.

> **SAY IT IN THE PPT**
>
> Nobody is left out: if a patient cannot use the kiosk, a nurse registers them with the same consent, safety checks and fair queue.

Reports: how the OPD is doing



Last 30 days of the demo hospital: eight cards, the daily trend, languages and department load.

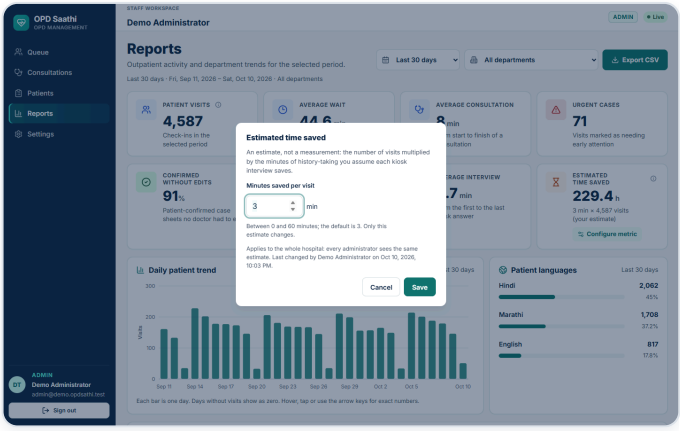
What every number means

Filters at the top — **Today, 7, 30 or 90 days** and **department** — control every card, the chart and the CSV export, so all numbers describe the same period.

CARD	MEANING	DEMO VALUE (30 DAYS)
Patient visits	Every check-in in the period, including patients not yet seen. A patient who comes twice counts twice.	4,580
Average wait	From joining the queue to the consultation starting.	44.6 min
Average consultation	From the start to the finish of a consultation.	8 min
Urgent cases	Visits marked as needing early attention.	71
Confirmed without edits	Of the case sheets patients confirmed, the share no doctor needed to change — a measure of how accurate the kiosk history is.	91%
Doctor edit rate	Of visits that produced a case sheet, the share a doctor corrected.	8.7%
Average interview	From the first to the last kiosk answer.	4.7 min
Estimated time saved	Visits × the minutes of history-taking each kiosk interview is assumed to save. An estimate, clearly labelled.	229 h (3 min × 4,580)

Charts and tables

- **Daily patient trend**: one bar per day with a labelled axis; days without visits show as zero; hover, tap or arrow keys show the exact number. Its total always equals the visits card.
- **Patient languages**: count and share of Hindi, Marathi and English.
- **Department load**: visits, urgent cases, average wait and share per department.
- **Export CSV** of the same period and department.
- "No data" (nothing to measure) is shown differently from a real 0.

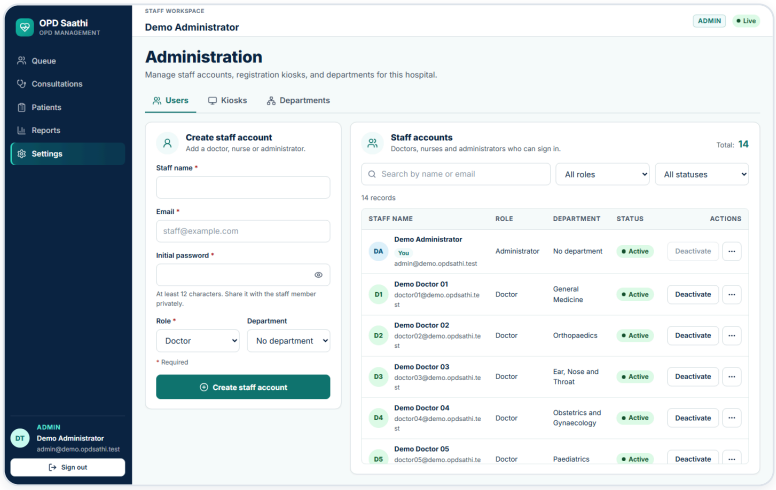


"Configure metric": one hospital-wide time-saved setting (0–60 minutes, default 3). Every change records who made it and when.

SAY IT IN THE PPT

Administrators see waiting times, consultation length, urgent cases, languages and department load for any period — and how much doctor time the kiosk saves.

Administration: users, kiosks, departments



Users: create an account on the left; search, filter and manage all staff on the right. The heading and tabs stay put while the list scrolls.

Users

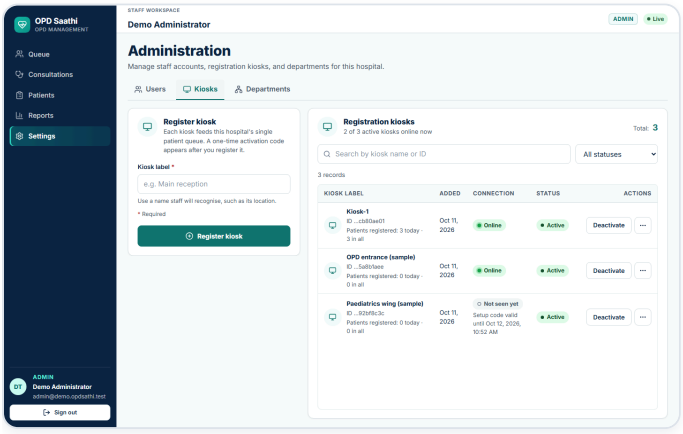
- Create doctors, nurses and administrators (name, email, initial password of 12+ characters, role, department).
- **Deactivate** asks first; the person is signed out within 10 minutes and can be activated again.
- ⋯ → **Delete** removes the account for good; their past work stays in patient history and the email can be used again.
- You can never deactivate or delete yourself.

Kiosks

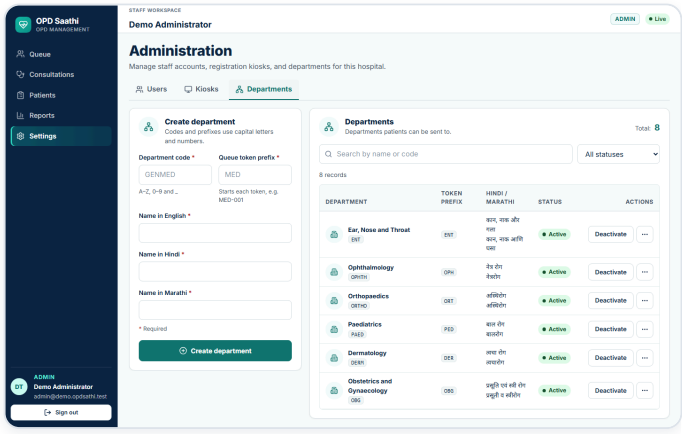
- Register → one-time code with **Copy code** and **Copy setup link**; valid for 24 hours.
- Connection: **Online**, **Offline**, Not seen yet, Turned off; "Setup code valid until …" while waiting.
- "Patients registered: N today · M in all" for each kiosk.
- ⋯ **menu**: New activation code, Delete.

Departments

- Code, token prefix (MED → MED-016), and names in English, Hindi and Marathi.
- **Deactivate** stops new patients at once.
- ⋯ → **Delete** only once no visit is still open there and no staff member is assigned to it.

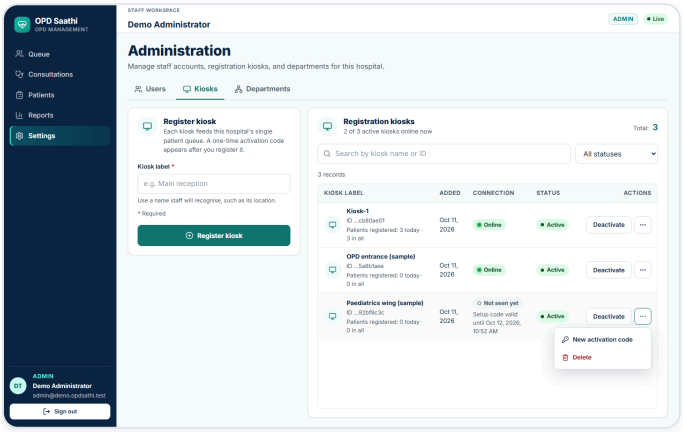


Kiosks: live connection, patients registered, and one kiosk still waiting for setup.

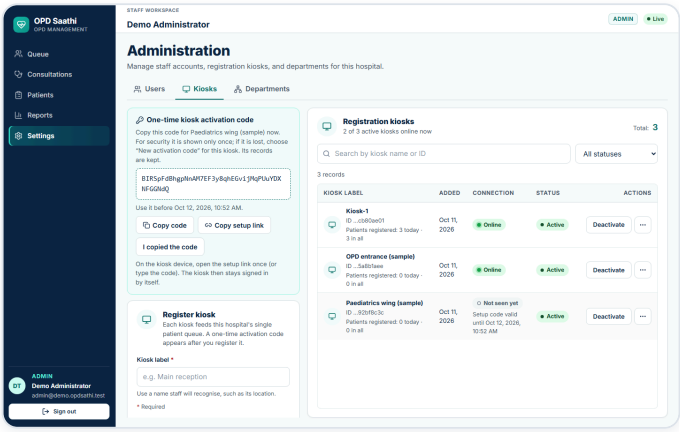


Departments with token prefixes and Hindi/Marathi names.

Kiosk life-cycle and safe delete

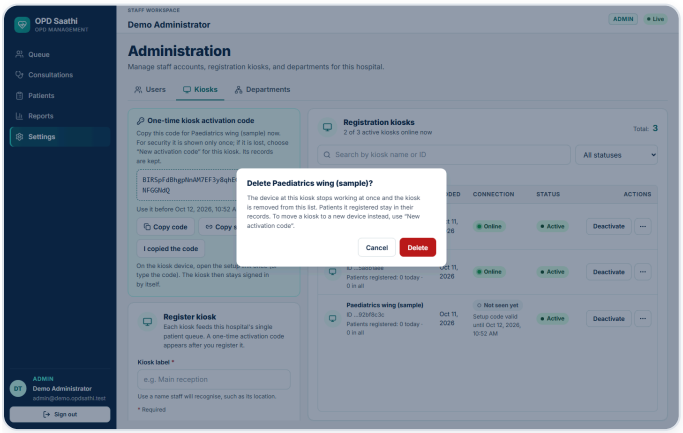


Every row has a ⋯ menu. For kiosks: New activation code and Delete.

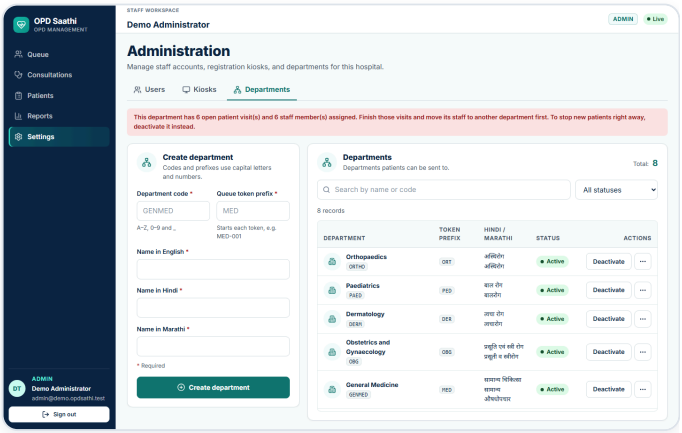


The one-time code with Copy code and Copy setup link, and the time it must be used by.

ACTION	WHAT HAPPENS	TO UNDO
Deactivate (turn off)	The kiosk shows "This kiosk is turned off" and cannot start check-ins. It stays paired to its device.	**Activate** — works at once, no new code needed.
New activation code	A fresh 24-hour code for the *same* kiosk record. The current device keeps working until the new code is used; then the new device takes over and the old one is signed out. History and "patients registered" are kept.	Not needed — the old code simply stops working.
Delete	The device stops working immediately and the kiosk leaves the list. Patients it registered stay in their records.	Cannot be undone from the screen — register a new kiosk instead.



Delete always asks first and says exactly what will happen.



A department that still has open visits or staff cannot be deleted; the message says what to do instead.

Delete never erases history. Visits, case sheets and the audit log keep pointing at the deleted staff member, kiosk or department, so medical records stay complete. The record just disappears from Settings and from every choice list, and its email, department code or token prefix can be used again.

Step-by-step administration tasks

Everything below is done by a signed-in administrator in **Settings** — no commands needed.

Add a department

1. **Departments** tab → *Create department*.
2. **Department code**: capital letters, numbers and _ (for example `CARDIO`).
3. **Queue token prefix**: up to 8 capitals/numbers (`CAR` gives tokens CAR-001, CAR-002…).
4. Name in English, Hindi and Marathi → **Create department**.

Automatic routing knows the standard codes (GENMED, ORTHO, ENT, OBG, PAED, DERM, OPHTH, SURG); other departments receive patients sent by staff.

Add a doctor, nurse or administrator

1. **Users** tab → *Create staff account*.
2. Name, email and an initial password of at least 12 characters.
3. Role and department (doctors and nurses see their department's queue).
4. **Create staff account**, then give the password to the person privately.

Set up a kiosk

1. **Kiosks** tab → type a label staff will recognise (*OPD entrance*, *Paediatrics wing*) → **Register kiosk**.
2. Click **Copy setup link**.
3. On the kiosk device, open the link once in Chrome or Edge (press F11 for full screen).
4. The kiosk shows *Start patient check-in* and, within a minute, **Online** in Settings.

For a kiosk on another device, open Settings with the server's network address (e.g. `http://192.168.1.20:5173/staff`), not `localhost`, before copying.

Replace a kiosk device, or after clearing its browser

1. Kiosks → ⋯ on that kiosk → **New activation code**.
2. Copy setup link → open it on the (new) device.

Same kiosk, same history. Do not register a new kiosk for this.

Turn something off, or delete it

- **Deactivate** = temporary (staff on leave, kiosk under repair, department closed today). **Activate** brings it back.
- ⋯ → **Delete** = permanent (person left, kiosk removed, department closed for good).
- A department can be deleted only after its open visits are finished and its staff are moved. Unfinished kiosk check-ins close by themselves overnight.

Staff changed department, or forgot their password

There is no edit screen yet. Delete the account and create it again with the same email and the new department or a new password — their past work stays in patient history. If the *only* administrator is locked out, use `create-admin` (see Command reference) to add another one.

12 · SCALE

One system, many hospitals

Every hospital gets its own OPD Saathi: its own departments, staff, kiosks, patients and queue. Nothing is shared between hospitals, and each hospital's administrator controls only their own.

District Hospital, Nagpur	GMC Hospital, Pune	PHC, Wardha
Own server and database	Own server and database	Own server — or shares a district server
12 kiosks → its own department queues	30 kiosks → its own department queues	2 kiosks → its own queues
Own administrator, doctors, nurses	Own administrator, doctors, nurses	Own administrator, doctors, nurses

Recommended: one installation per hospital

- OPD Saathi runs on a PC or server **inside the hospital**, with its own PostgreSQL database.
- Patient data stays in the building, and check-in and the queue keep working if the internet goes down (AI features need internet).
- Each installation has its own passwords and encryption keys.
- Installed with one script: `.\setup-hospital.ps1` (next section).

Also possible: several hospitals on one server

- For example a district hosting its PHCs centrally. Add each hospital with `.\setup-hospital.ps1 -AddHospital`.
- Staff choose their hospital on the sign-in page; each sees only their own hospital.
- Separation is enforced **by the database itself** (row-level security), not just by the app.

How many kiosks?

As many as the hospital needs. Every kiosk belongs to exactly one hospital and sends its patients into that hospital's department queues — 100 kiosks still mean **one** General Medicine queue, one Orthopaedics queue, and so on, in fair calling order. Tokens are numbered per department per day (MED-001, MED-002…) whichever kiosk the patient used. Each kiosk shows its own hospital's name on screen and on the printed token.

Tested

A second hospital was set up with the new command and checked live: its patient appeared only in its own queue, and neither hospital could open, call, change or delete the other's patients, staff or kiosks (20 of 20 checks passed).

> **SAY IT IN THE PPT**
> Each hospital gets its own OPD Saathi — its own data, kiosks and staff — and the database itself guarantees that one hospital can never see another's patients.

13 · INSTALLATION

Setting up a new hospital

For the person installing OPD Saathi on site. This is separate from the demo — never use `start-demo.ps1` for a real hospital.

You need

- A Windows PC or server on the hospital network that stays on during OPD hours.
- **Docker Desktop** installed and running (set it to start with Windows).
- A copy of the OPD Saathi folder.
- Kiosk devices (tablet or PC) and staff PCs with Chrome or Edge, on the same network.
- The hospital's name, a short code for it, and the first administrator's name and email.

What the setup script does

1. Creates `.env` with fresh random passwords and encryption keys for this hospital.
2. Builds and starts the database, cache, API, worker and web app; applies database updates.
3. Asks for the hospital and its first administrator, then creates them with the 8 standard departments.
4. Prints the address staff PCs and kiosks use.

Install (PowerShell, in the OPD Saathi folder)

```
cd C:\OPD-Saathi
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
.\setup-hospital.ps1
```

It asks:

```
Hospital name (shown on its kiosks): District Hospital, Nagpur
Short unique hospital code (e.g. gmc-nagpur): dh-nagpur
Administrator's full name: Dr. A. Sharma
Administrator's email: admin@dhnagpur.example
Administrator password (12+ characters): ************
Type it again: ************
```

and finishes with:

```
Hospital ready: District Hospital, Nagpur (019b…)
Administrator: admin@dhnagpur.example
Departments: 8 standard departments created.
  Staff (any PC on the hospital network):  http://192.168.1.20:5173/staff
  Kiosks:                                  http://192.168.1.20:5173/kiosk
```

Then, the same day

1. The administrator signs in at the **Staff** address and opens Settings.
2. **Departments**: keep, rename, deactivate or delete the standard ones; add any others.
3. **Users**: add every doctor and nurse with their department.
4. **Kiosks**: register each kiosk and open its setup link once on that device.
5. Optional: put `GEMINI_API_KEY` (and `SARVAM_API_KEY`) in `.env` and run `.\setup-hospital.ps1 -NoBuild` to switch on AI and natural voices.
6. Take the first backup (see Command reference) and store `.env` safely with it.

Options and configuration

Setup script options

COMMAND	USE IT TO
`.\setup-hospital.ps1`	First installation. Safe to run again: it restarts or updates the installation and never replaces existing database passwords or encryption keys.
`.\setup-hospital.ps1 -NoBuild`	Restart quickly without rebuilding (after editing `.env`, or if something stopped).
`.\setup-hospital.ps1 -AddHospital`	Add another hospital to this same server.
`.\setup-hospital.ps1 -WebPort 8080`	Use another port if 5173 is taken.

On a Linux or Mac server (no PowerShell)

```
cp .env.example .env      # replace every "replace-…" value:
                          #   passwords and JWT_SECRET: openssl rand -hex 32
                          #   APP_DATA_KEY, APP_HMAC_KEY, CURSOR_HMAC_KEY: openssl rand -base64 32
                          #   add WEB_BIND=0.0.0.0, WEB_ORIGIN and PUBLIC_WEB_URL=http://<server-ip>:5173
docker compose up -d postgres redis redis-cache worker-role-bootstrap
docker compose run --rm api alembic -c /app/alembic.ini upgrade head
docker compose up -d api worker web
docker compose run --rm api python -m app.cli setup-hospital
```

The `.env` file

SETTING	WHAT IT IS
`APP_DATA_KEY`, `APP_HMAC_KEY`	Encrypt patient names, phones and answers. **Never change them** after patients exist, and back them up — without them the data cannot be read.
`JWT_SECRET`, `CURSOR_HMAC_KEY`	Sign staff and kiosk sign-ins. Changing them signs everyone out once (kiosks then need a new activation code).
`POSTGRES_PASSWORD`, `OSA_*_PASSWORD`, `REDIS_PASSWORD`	Database and cache passwords, created by the script. Do not change after installation.
`WEB_PORT`, `WEB_BIND`, `WEB_ORIGIN`, `PUBLIC_WEB_URL`	The address and port staff, kiosks and patients' phones use. Set by the script from the server's network address.
`GEMINI_API_KEY`, `SARVAM_API_KEY`	Optional AI and voice keys. Leave empty to run on rules and the browser's voice.
`NOTIFICATION_PROVIDER` and `TWILIO_*` / `WHATSAPP_*`	Optional SMS/WhatsApp queue messages (off by default; `console` only logs them).
`DEMO_*`	Used only by the demo script.

Running it day to day

Keeping it running

- All services restart by themselves when Docker Desktop starts — set Docker Desktop to start with Windows.
- `docker compose down` stops everything and **keeps** the data.
- **Never** run `docker compose down -v` on a hospital server: it deletes the database.
- Updating: replace the OPD Saathi folder with the new version (or `git pull`), then run `.\setup-hospital.ps1`.

Before going live

- **Voice on kiosks** needs HTTPS: browsers allow the microphone only on `https://` or on the server itself. Until a certificate is set up, network kiosks can tap and type but not speak — or start Chrome on managed kiosks with `--unsafely-treat-insecure-origin-as-secure=http://<server-ip>:5173`.
- Back up the database and `.env` together, regularly.
- Keep `.env` private — it holds this hospital's keys.

Updating to a new version

1. Take a backup (`sh scripts/backup.sh`) and keep a copy of `.env`.
2. Replace the OPD Saathi folder with the new version, keeping the existing `.env` (or run `git pull`).
3. Run `.\setup-hospital.ps1`. It rebuilds, applies database updates and restarts. Data, staff and kiosks are kept; kiosks stay signed in.
4. On each kiosk and staff PC, press Ctrl+Shift+R once to load the new version.

A typical morning

- The server and Docker Desktop are already on; kiosks show *Start patient check-in*.
- The administrator glances at Settings → Kiosks: "N of M active kiosks online now". An Offline kiosk usually just needs its device switched on or reconnected to the network.
- Unfinished visits from the previous day were closed automatically overnight, so every queue starts empty.

> **SAY IT IN THE PPT**
> Installing OPD Saathi at a new hospital is one script; after that the hospital runs it from the Settings page, and updates keep all data and kiosks in place.

AI features — and life without AI keys

AI makes the system smarter, but it is optional. Without any keys, every core feature still works using rules, the question bank and the browser's own voice.

FEATURE	WITH AI (GEMINI; SARVAM OPTIONAL FOR VOICE)	WITHOUT ANY KEYS
English case sheet	Every Hindi/Marathi answer translated, plus a 2–3 line doctor's brief	Tapped answers translated from the bilingual question bank; spoken/typed words stay in the patient's language
Question order	AI picks the most useful next question and can ask a short follow-up (labelled "AI follow-up")	Fixed, logical question flow
Voice answers	Server speech recognition (Sarvam for the best Hindi/Marathi, else Gemini)	Browser speech recognition (Chrome/Edge)
Read aloud	Natural Indian voices	Browser voice
Old prescription photo	Medicines read automatically, patient confirms	Button hidden
Safety alerts	Rules + AI second opinion (can only raise urgency and must quote the patient)	Rules engine
Triage check	AI review after check-in; "AI suggests… please confirm" alert	Score from rules
Routing, queue, tokens, reports	Always rules-based and work the same with or without AI	

Guard-rails on every AI answer

- An output guard rejects any AI text that diagnoses ("you have…"), advises treatment, names doses or contains links.
- AI red flags must quote what the patient actually said.
- AI summaries are labelled "not verified by staff" until a person checks them.
- If an AI service fails or is slow, the kiosk quietly continues with the question bank.

Why this design

Hospitals can start with zero AI cost and zero data leaving the building, then turn AI on when they approve it. Safety never depends on AI being right — rules decide, AI only adds caution, and people confirm.

SAY IT IN THE PPT

AI is an assistant, not the authority: it translates, summarises and double-checks, but it can never diagnose, never lower urgency, and the whole system still runs if it is switched off.

Safety, privacy and security

Clinical safety

- History only — no diagnosis, disease names, medicines or advice, anywhere.
- Danger signs first; uncertainty always escalates, never lowers urgency.
- The patient confirms the read-back; doctors can correct anything; every change is versioned.
- 15 red-flag rules and the triage weights are readable drafts awaiting clinician sign-off.

Privacy

- Names, phone numbers, interview answers and case sheets are **encrypted** in the database.
- Minimal data: only age and sex are required; phone is optional.
- The phone status page shows no personal or health details.
- The offline kiosk form is encrypted on the device and deleted after sync.
- Optional data-retention tool removes old transcripts and pseudonymises patients.

Security

- Each hospital's data is walled off inside the database (row-level security), not just in code.
- Roles: admin, doctor, nurse, kiosk, patient — each sees only what it needs.
- Passwords hashed; repeated failed sign-ins lock the account and are rate-limited.
- Kiosks are set up with one-time codes that expire after 24 hours, then sign in with their own device key; deleting or re-pairing a kiosk cuts it off at once.
- Deactivated or deleted staff lose access within 10 minutes; nobody can lock themselves out.
- Each installation signs sign-ins with its own private key; the setup scripts never use the published defaults.
- The web app sends strict browser security headers (content security policy, no framing), so injected scripts cannot run.
- Every important action goes to an append-only audit log.

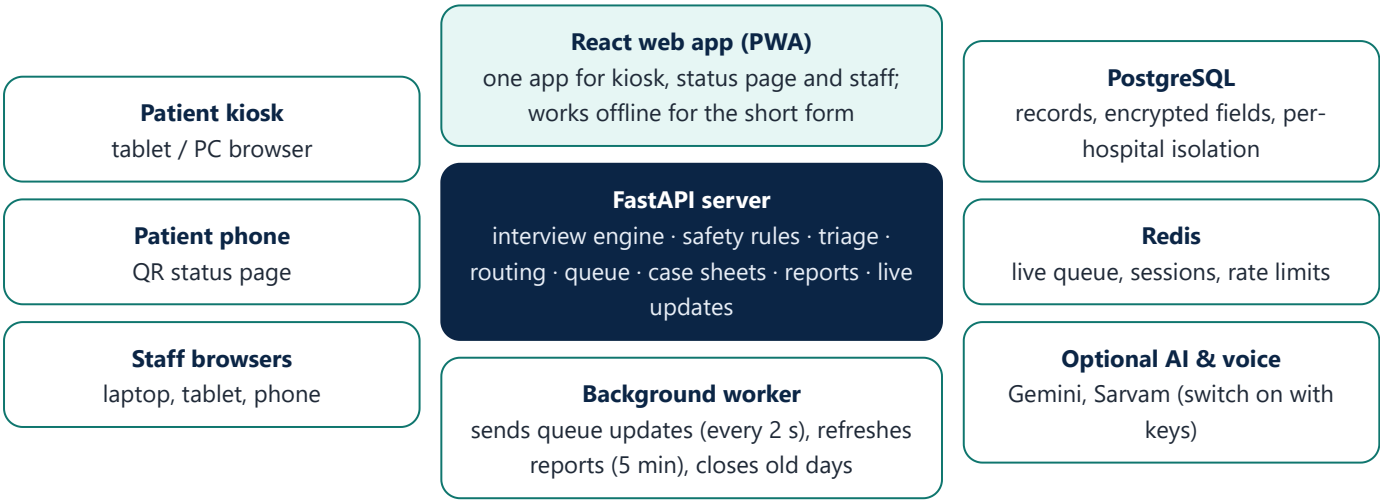
Reliability

- Requests carry a unique key, so a double tap or a retried network call never creates two visits.
- Two staff editing one case sheet cannot overwrite each other.
- CSV exports are protected against spreadsheet formula tricks.
- Automated security checks run on every change (dependency audit, code scanning, secret scanning).

SAY IT IN THE PPT

Built for health data from day one: encrypted records, each hospital's data isolated inside the database, role-based access, audit logs, and AI that can never give medical advice.

How it is built



LAYER	TECHNOLOGY
Web app	React 18 + TypeScript, Vite, installable PWA, i18next (Hindi/Marathi/English), TanStack Query, virtualised lists for very long queues
API	Python 3.12, FastAPI, Pydantic, SQLAlchemy, WebSockets for live queue updates, PDF generation for case sheets
Data	PostgreSQL with row-level security and migrations (Alembic); Redis for queue state and a separate cache
Running it	Docker Compose (database, Redis, API, worker, web); `setup-hospital.ps1` installs a real hospital, `start-demo.ps1` a complete demo hospital
Quality gates	GitHub Actions on every change: backend, frontend and security jobs must all pass before code reaches the main branch

SAY IT IN THE PPT

A modern, open stack: a single React web app for every screen, a Python FastAPI server with rules-based safety at its core, PostgreSQL and Redis — all started with one command.

17 · EVIDENCE

Quality, measured

814
backend unit tests
plus database integration tests in CI

132
browser tests
real screens at 7 widths, 3 languages, accessibility scans

54
web unit tests
queue, case sheet, reports, settings, kiosk

Synthetic evaluations (from the project's evaluation report)

CHECK	RESULT
Interview scenarios completed correctly	30 / 30
Department routing test cases	96 / 96
Red flags that must be caught	60 / 60
Negations not falsely flagged ("no chest pain")	50 / 50
History extraction fixtures with zero invented fields	100 / 100
Adversarial prompts blocked by the AI output guard	100 / 100
Urgency never decreases when a warning sign is added	500 / 500

Usability and performance

- Accessibility: zero serious or critical issues in automated scans; Lighthouse accessibility score 100 on kiosk and status pages; all colour pairs pass WCAG AA contrast.
- No sideways scrolling on any staff page at 1600, 1440, 1366, 1024, 768, 390 and 320 px wide.
- A 5,000-patient queue applies an urgent update in well under 100 ms.
- Kiosk touch targets are at least 64 × 64 px.

> These are synthetic tests: they prove the software behaves as designed, not clinical accuracy. Clinician and native-speaker review is still required (see limitations).

Running the demo and the 3-minute script

Start everything

On any Windows, Mac or Linux machine with Docker: `powershell -ExecutionPolicy Bypass -File .\start-demo.ps1`. The script creates secrets, starts all services, builds a fictional "Demo Government Hospital, Nagpur" with 8 departments, one doctor per department, five General Medicine nurses and an administrator, adds a realistic month of past OPD days for Reports, and prints a one-click kiosk link plus staff logins. Add `-NoBuild` to start faster when the code has not changed. Put a `GEMINI_API_KEY` (and optionally `SARVAM_API_KEY`) in `.env` to turn on AI and natural voices.

DEMO LOGIN	EMAIL	WORKS IN
Administrator	`admin@demo.opdsathi.test`	Everything, including Reports and Settings
Doctors (8)	`doctor01` … `doctor08@demo.opdsathi.test`	01 General Medicine · 02 Orthopaedics · 03 ENT · 04 Obstetrics & Gynaecology · 05 Paediatrics · 06 Dermatology · 07 Ophthalmology · 08 General Surgery
Nurses (5)	`nurse01` … `nurse05@demo.opdsathi.test`	General Medicine, with doctor01

Passwords are printed by the script and kept in `.env` (`DEMO_ADMIN_PASSWORD`, `DEMO_STAFF_PASSWORD`); they are deliberately not written in this guide.

Screen setup: kiosk full-screen in one window, staff workspace (signed in as doctor01 or admin) in another, a phone (or phone-sized window) for the QR.

TIME	SHOW	SAY
0:00	Kiosk welcome; tap मराठी, then Start.	"A patient picks their language; everything from here is in Marathi."
0:15	Consent is read aloud; I agree; age and sex.	"Consent first. Only age and sex are needed."
0:35	Safety check: "No, none of these".	"Danger signs come first, so a chest-pain patient alerts the nurse within seconds."
0:45	Tap fever and cough; answer follow-ups; speak one answer.	"Most answers are one tap, or just speak. Questions adapt to the complaint."
1:30	(With AI) scan an old prescription on the medicines question.	"Old paper prescriptions are read automatically; the patient confirms."
1:45	Review is read back; Yes, this is correct.	"The patient hears and confirms their history before it reaches the doctor."
2:00	Token, department, place in line, QR; scan with a phone.	"They follow their turn on their own phone."
2:15	Staff: queue shows the new token in English; click Why?; open the case sheet.	"A fair, explainable queue, and a one-page English case sheet with the patient's own words."
2:45	New check-in, tap Chest discomfort → urgent alert; press Call staff → help alert.	"Urgent cases and patients who need help reach staff in real time."

Command reference

Run these in PowerShell (or a terminal) inside the OPD Saathi folder. Day-to-day work needs none of them — administrators use Settings.

Start, stop and check

COMMAND	WHAT IT DOES
`.\setup-hospital.ps1`	Install or restart a real hospital (options: `-NoBuild`, `-AddHospital`, `-WebPort 8080`).
`.\start-demo.ps1`	Start the demo hospital (options: `-NoBuild`, `-WebPort 8080`).
`docker compose ps`	Show which services are running and healthy.
`docker compose logs -f api`	Watch the server's log (`worker` or `web` for the others). Ctrl+C to stop watching.
`docker compose down`	Stop everything. Data is kept.
`docker compose run --rm api alembic -c /app/alembic.ini upgrade head`	Apply database updates (the scripts do this for you).
`Invoke-RestMethod http://localhost:5173/api/auth/hospitals`	List the hospitals on this server with their IDs.

Hospitals, administrators and kiosks

Each line starts with `docker compose run --rm api python -m app.cli`, shortened here to `cli`.

COMMAND	WHAT IT DOES
`cli setup-hospital`	Create a hospital, its first administrator and the 8 standard departments. Asks for anything not given: `--name "…" --code dh-nagpur --admin-name "…" --admin-email …`. Add `--no-standard-departments` to start empty. In scripts, pass the password with `-e ADMIN_PASSWORD=…` after `run`.
`cli create-admin --hospital-id <id>`	Add another administrator to an existing hospital (asks name, email and password) — the way back in if the only administrator is locked out.
`cli create-kiosk --hospital-id <id> --label "OPD entrance"`	Register a kiosk without the Settings page; prints its one-time code.
`cli queue-rebuild`	Rebuild the live queue from the database (only if the cache was lost).

Demo data, backups and retention

As before, `cli` stands for `docker compose run --rm api python -m app.cli`.

Demo data

COMMAND	WHAT IT DOES
`cli seed-demo`	Create or refresh the demo hospital, its departments and the demo staff.
`cli demo-kiosk`	Print a setup link for the demo hospital's "Demo kiosk" (re-uses the same kiosk each time).
`cli seed-demo-data` / `cli clear-demo-data`	Add / remove five sample visits in today's queue.
`cli seed-demo-history --days 30` / `cli clear-demo-history`	Add / remove a realistic month (1–120 days) of finished OPD days for Reports.
`cli seed-bulk --visits 100000` / `cli clear-bulk`	Add / remove load-test visits.

Backups and data retention

COMMAND	WHAT IT DOES
`sh scripts/backup.sh`	Back up the database and queue state to `backups/<date-time>/` with checksums (Git Bash or Linux). Store `.env` with it.
`RESTORE_DATABASE=osa_restore_test sh scripts/restore.sh backups/<date-time>`	Check a backup by restoring it into a separate database and verifying the visit count. It never overwrites the live database.
`make retention HOSPITAL_ID=<id>` / `make retention-apply HOSPITAL_ID=<id>`	Show / apply the retention policy: remove interview records older than 30 days and pseudonymise patients after 90 (change with `RETENTION_DAYS`, `ANONYMISE_DAYS`). Needs the project's Python environment.

Troubleshooting

YOU SEE	WHY, AND WHAT TO DO
Kiosk shows the activation screen	The kiosk was deleted, set up on another device, or its browser data was cleared. Settings → Kiosks → ⋯ → **New activation code** → open the setup link on the kiosk.
"The activation code is invalid, expired, or has already been used"	Codes work once and for 24 hours. Issue a new activation code.
Setup link opens a blank or wrong page on another device	It was copied from a `localhost` admin page. Open Settings with the server's network address and copy the link again.
Kiosk: "This kiosk is turned off"	An administrator deactivated it. Settings → Kiosks → **Activate**. No new code needed.
Kiosk: "Ask staff to check this kiosk"	The hospital has no active department. Activate at least one (normally General Medicine).
Microphone or "Tap and speak" does not work on a network kiosk	Browsers allow the microphone only over HTTPS. Set up a certificate, or use the Chrome option in section 13. Tapping and typing still work.
Old version still shows after an update	Press Ctrl+Shift+R (the app keeps an offline copy). Reload once more if needed.
Staff: "account locked"	Five wrong passwords lock an account for 15 minutes. Wait, or recreate the account (see Admin how-to).
"Department still in use" when deleting	Finish its open visits and move its staff first, or deactivate it instead. Unfinished kiosk check-ins close by themselves overnight.
"Already exists" when creating	That email, department code or token prefix is already used in this hospital. Choose another, or delete the old record first.
Page will not load / "Cannot connect to the Docker daemon"	Start Docker Desktop, wait until it says "Engine running", then run the setup (or demo) script again with `-NoBuild`.
Port 5173 is in use	The scripts switch to 8080, or choose one with `-WebPort`.

21 · HONEST STATUS

Limitations and next steps

Being clear about what is not done yet makes the presentation stronger, not weaker.

ITEM	STATUS AND NEXT STEP
Clinical review	Question banks, red-flag rules, triage weights and consent text are drafts. They need sign-off by clinicians, and the Hindi and Marathi wording by native speakers.
Doctor's clinical notes	Examination findings, assessment and plan are not stored yet; doctors record them in the hospital's own system. Adding them needs a small database and API extension.
Production deployment	A hospital can now be installed with `setup-hospital.ps1` (own database, secrets, standard departments). Still to add: HTTPS on the hospital network (needed for kiosk voice), scheduled backups and monitoring.
Settings screens	No screen yet to edit a staff member's details or department, reset a password, switch AI on/off per hospital, or remove a whole hospital. The first three are supported by the server and need small screens; removing a hospital is an operator task.
SMS / WhatsApp	The opt-in and message pipeline exist; a provider account must be connected to actually send messages.
Hospital records integration	No HIS, ABHA/ABDM or FHIR connection yet — a natural next phase.
Offline voice	The offline short form works without internet; the full voice interview needs the local server.
Human testing	Still to do: run the full demo without internet, and have first-time patients complete check-in without coaching.

Natural roadmap

1. Clinician and language review of all medical content.
2. Pilot in one OPD department with real kiosks and staff feedback.
3. Doctor's notes and e-prescription fields; SMS provider.
4. ABHA/ABDM linking and hospital-system integration.
5. More languages (Tamil, Telugu, Bengali…) — the system is built for it.

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Suggested PPT outline

#	SLIDE	KEY POINTS	VISUAL
1	Title	OPD Saathi — multilingual check-in kiosk and live queue for government OPDs. Team, SIH26047.	Cover
2	The problem	Crowded OPDs, many languages, unstructured history, unknown waiting time, little doctor time.	Photo / numbers
3	Our solution in one line	History collected before the doctor, fair live queue, English case sheet, admin insight.	Journey flow (page 2)
4	Patient kiosk	3 languages; tap/speak/type; danger signs first; adaptive 8–15 questions; read-back; token + QR.	k5, k8 screenshots
5	Inclusive by design	Voice guidance, high contrast, 64 px buttons, call staff, offline form, assisted registration.	Kiosk + icons
6	Phone status	Live place in line and wait on the patient's own phone; no app.	Phone screenshot
7	Fair, explainable queue	Urgent first; 0.6 min per point, max 45; "10/10" can't cheat; "Why?" for every score.	Why? popover
8	Staff live queue	Counters, filters, alerts, call/start/complete, override with reason, next to call.	Queue screenshot
9	Doctor's case sheet	Red flags first, AI brief marked unverified, full history, patient's own words, versioned edits, PDF.	Consultation overview
10	AI — assistant, not authority	Translate, summarise, follow-ups, prescription reading, second opinion; output guard; works without AI.	AI table
11	Admin insight	Visits, waits, consultation time, urgent cases, languages, departments, time saved; users, kiosks, departments.	Reports screenshot
12	Safety, privacy, security	No diagnosis; encryption; per-hospital isolation; roles; audit; one-time kiosk codes.	Four cards
12b	Ready for many hospitals	One setup command per hospital; own database; many kiosks, one fair queue; set up a kiosk by opening one link.	Hospitals diagram (section 12)
13	Technology	React PWA, FastAPI, PostgreSQL, Redis, Docker; one-command setup.	Architecture diagram
14	Proof it works	814 + 54 + 132 tests; evaluation scores; accessibility 100; live demo.	Numbers
15	Impact and next steps	Minutes saved per patient, calmer waiting rooms, clinician review, pilot, ABHA, more languages.	Roadmap

> Tip: the two strongest live-demo moments are (1) tapping "Chest discomfort" on the kiosk and watching the urgent alert appear instantly in the staff queue, and (2) clicking "Why?" to show the transparent score.

Words used in the app

TERM	MEANING
Token	The patient's number for the day, made of the department prefix and a counter (MED-016 = 16th General Medicine patient today).
Danger tile	One of six large buttons on the kiosk safety screen for signs that need immediate attention.
Red flag	A warning raised by a safety rule (or a danger tile, or an AI second opinion) from what the patient reported. Not a diagnosis.
Triage score	0–100 number, staff-only, that decides how much head start a patient gets in the queue.
Routine / Priority / Urgent	Score below 30 / 30 or more / 70 or more or any urgent red flag. Urgent goes to the front.
Head start	How much earlier than their arrival time a patient is placed: 0.6 minutes per score point, at most 45 minutes.
Case sheet	The organised English history the doctor reads, with the patient's original words and the source of every field.
Read-back	The summary the kiosk reads to the patient for confirmation before check-in finishes.
Transcript	The unchangeable record of every question and the patient's exact answer.
Override	A staff change of priority or department, which requires a written reason and is recorded.
Assisted registration	A nurse checking in a patient who cannot use the kiosk, with the same consent and safety steps.
Activation code	A one-time secret, valid 24 hours, that turns a browser into a trusted kiosk for one hospital.
Setup link	The kiosk page address with the activation code attached; opening it once activates the kiosk.
Device key	The kiosk's own long-lived sign-in, received at activation, that keeps it signed in until it is deleted or re-paired.
Deactivate / Delete	Deactivate turns a record off and can be undone; Delete removes it from Settings for good but keeps all history that refers to it.
Hospital code	A short unique name for a hospital on a server (for example `dh-nagpur`), chosen at setup.
Heartbeat	The small "I'm on" signal each kiosk sends about once a minute, shown as Online/Offline in Settings.
PWA	Progressive Web App — a website that can be installed like an app and keeps working offline for the short form.
Row-level security	A database feature that makes each hospital's rows invisible to every other hospital.

OPD Saathi collects history for clinical staff. It does not diagnose or recommend treatment. All patients, staff and hospitals shown in this guide are fictional.